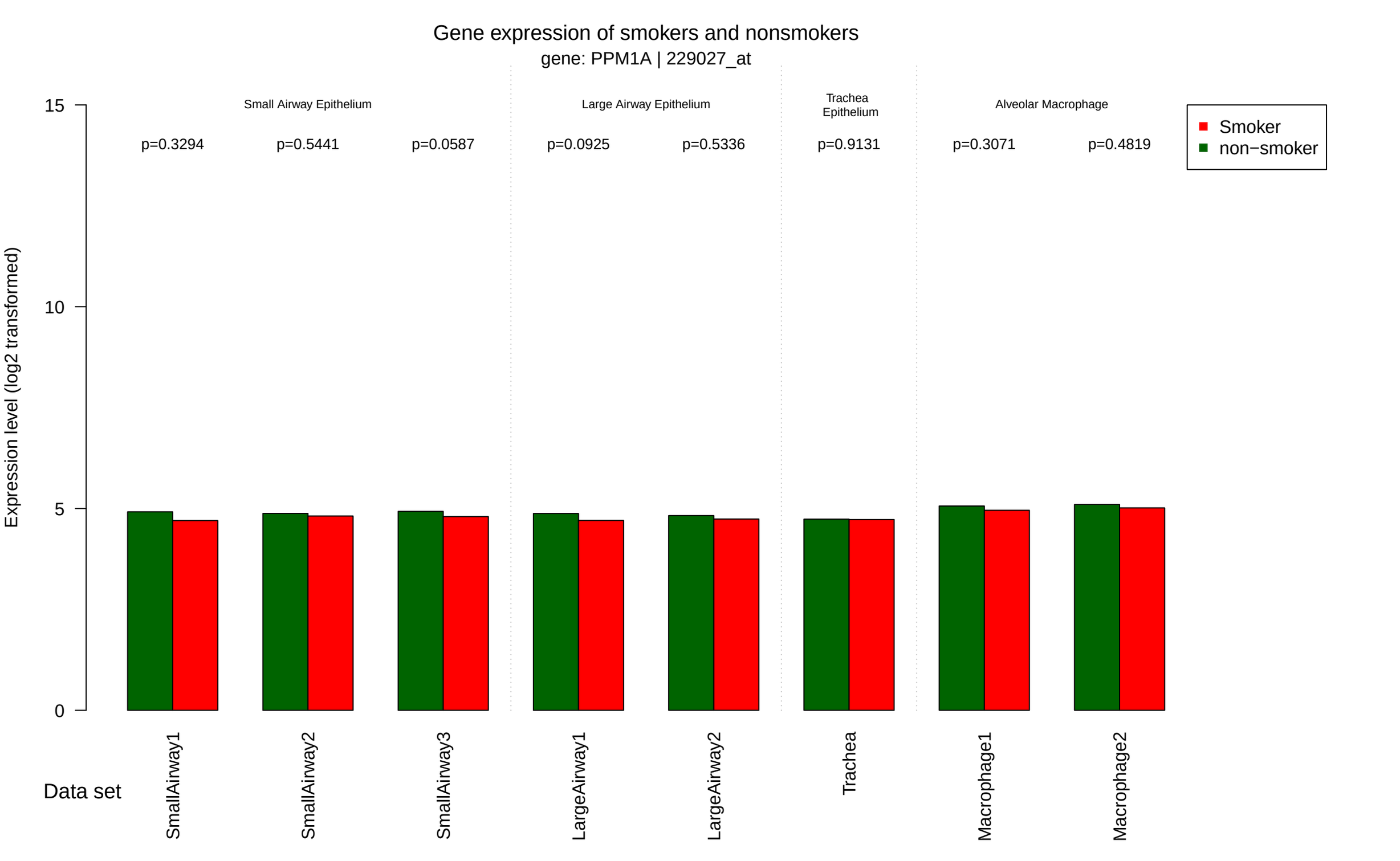
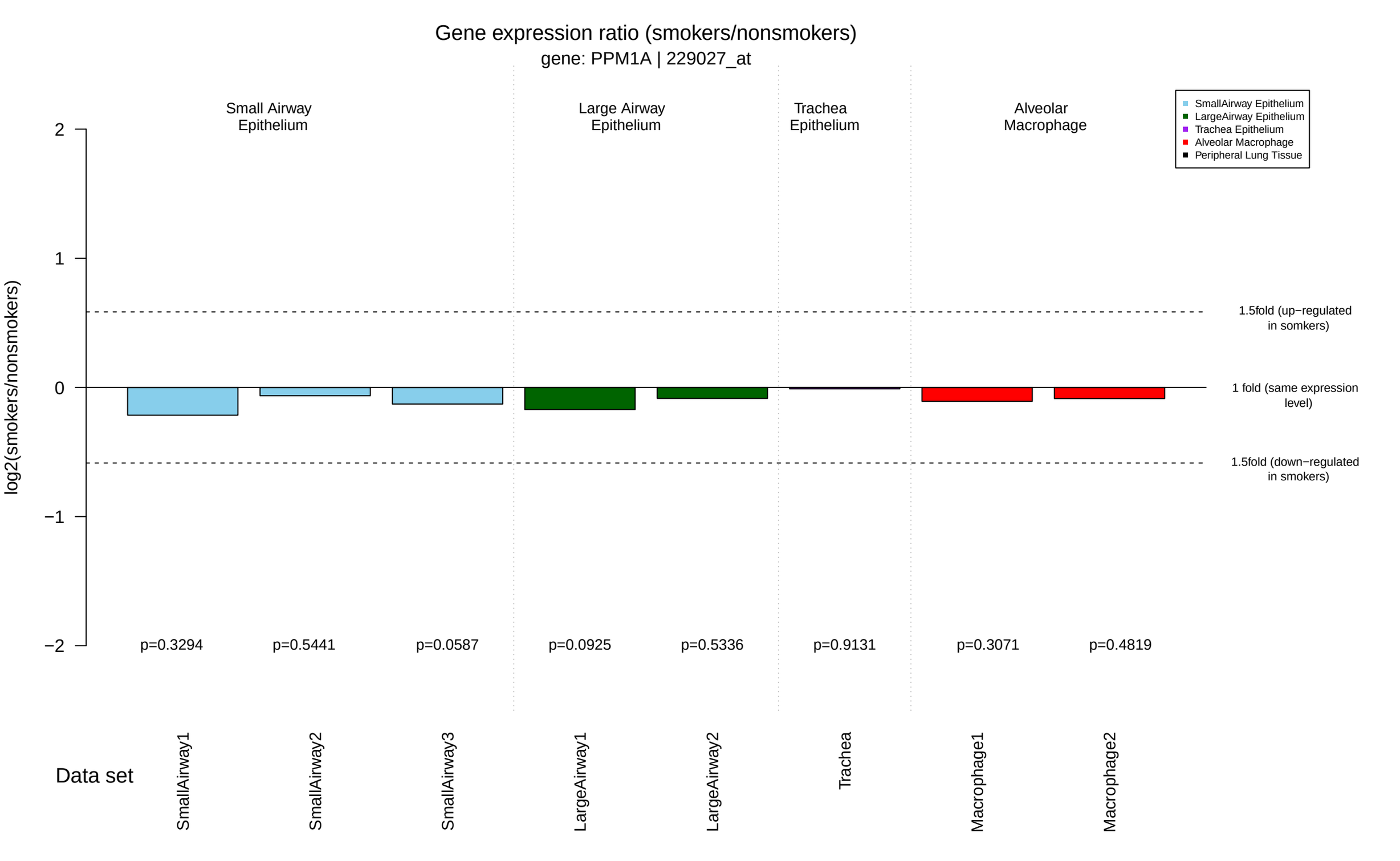
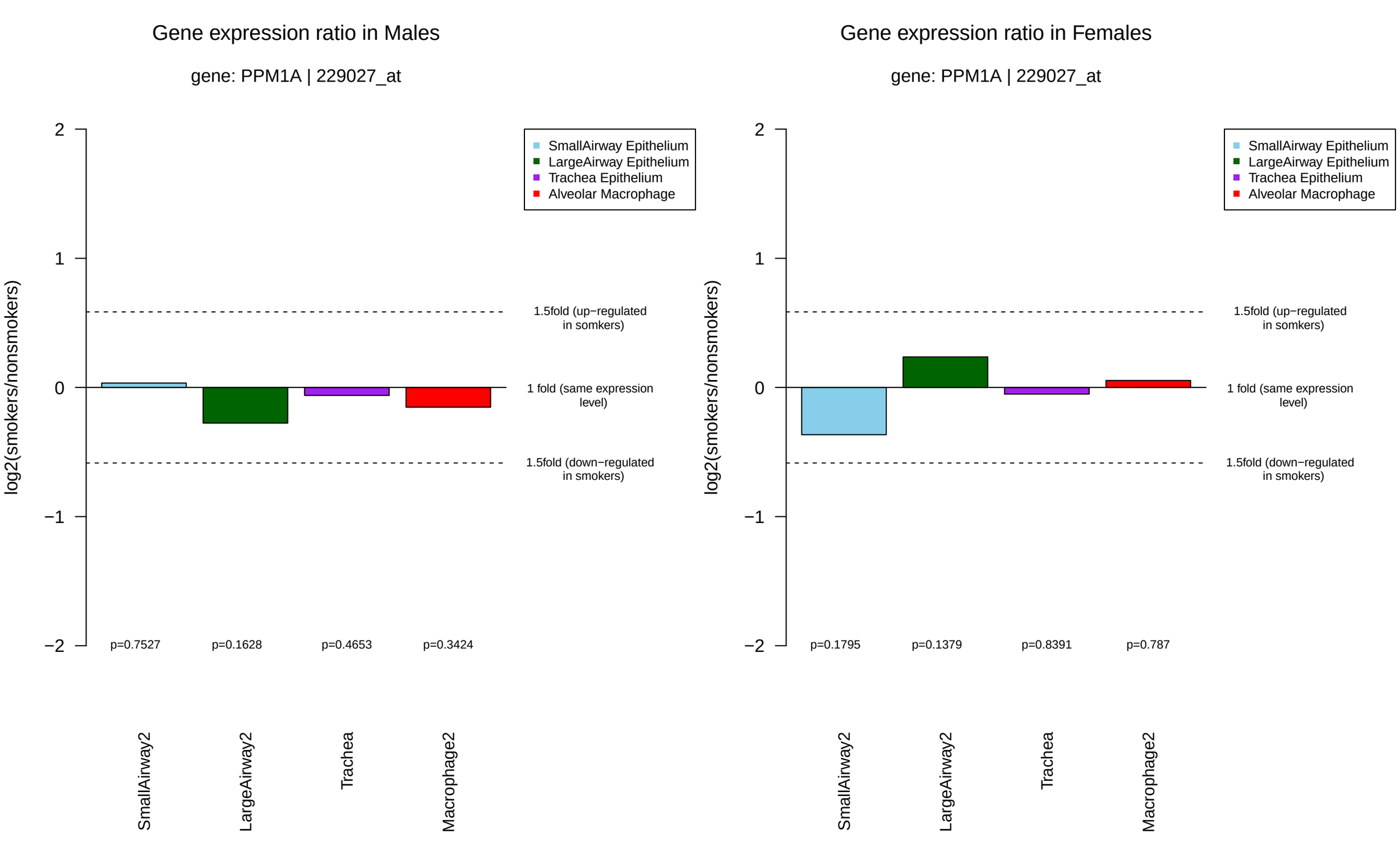
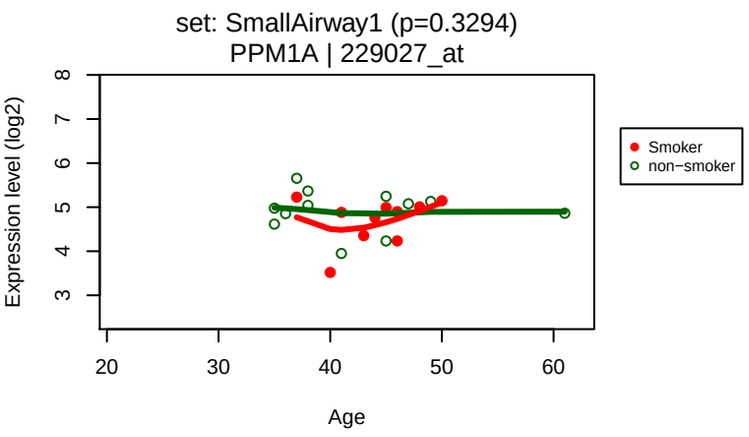
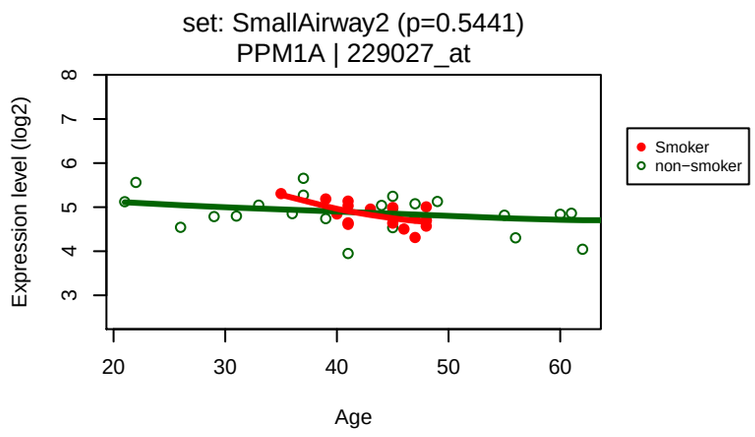
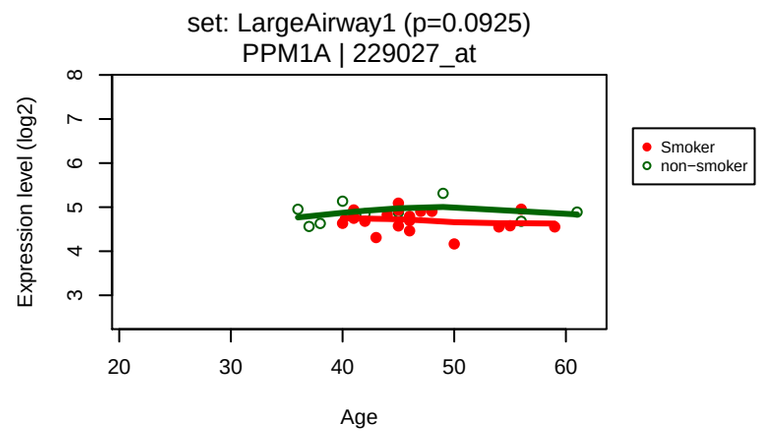

Gene expression of smokers and nonsmokers
gene: PPM1A | 229027_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3294
p=0.5441
p=0.0587
p=0.0925
p=0.5336
p=0.9131
p=0.3071
p=0.4819
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

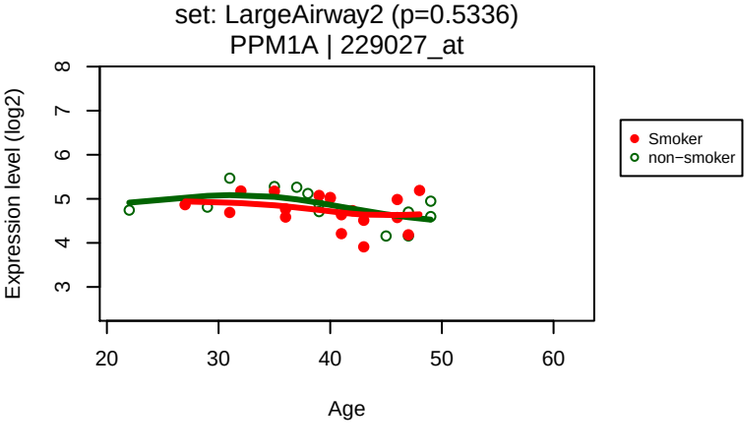

Gene expression ratio (smokers/nonsmokers)
gene: PPM1A | 229027_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3294
p=0.5441
p=0.0587
p=0.0925
p=0.5336
p=0.9131
p=0.3071
p=0.4819
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

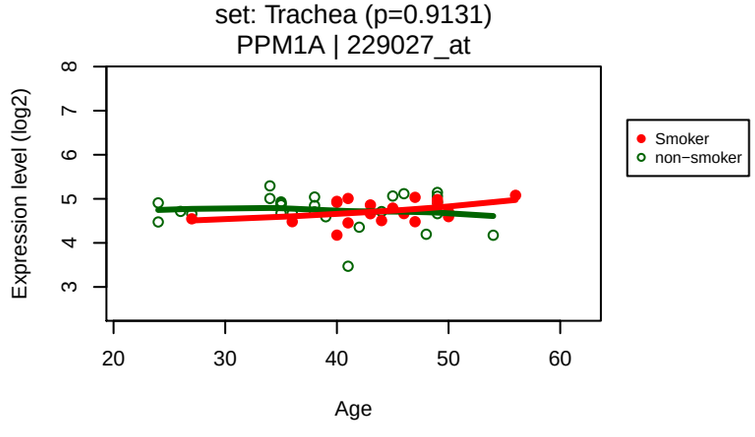

Gene expression ratio in Males
gene: PPM1A | 229027_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7527
p=0.1628
p=0.4653
p=0.3424
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PPM1A | 229027_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1795
p=0.1379
p=0.8391
p=0.787
SmallAirway2
LargeAirway2
Trachea
Macrophage2

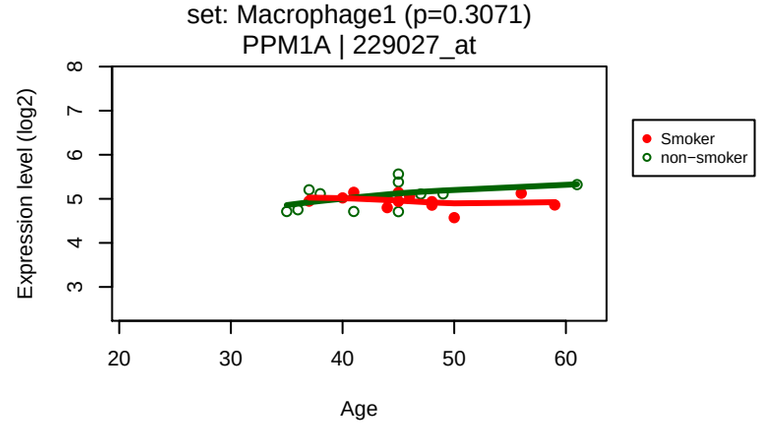

set: SmallAirway1 (p=0.3294)
PPM1A | 229027_at
Expression level (log2)
Age
Smoker
non−smoker

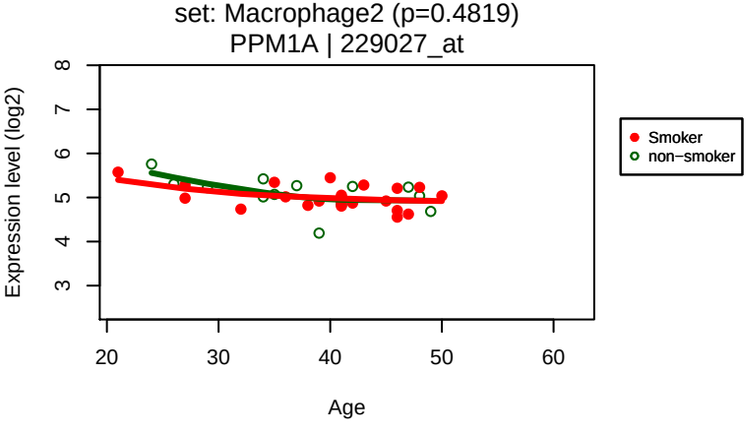

set: SmallAirway2 (p=0.5441)
PPM1A | 229027_at
Expression level (log2)
Age
Smoker
non−smoker

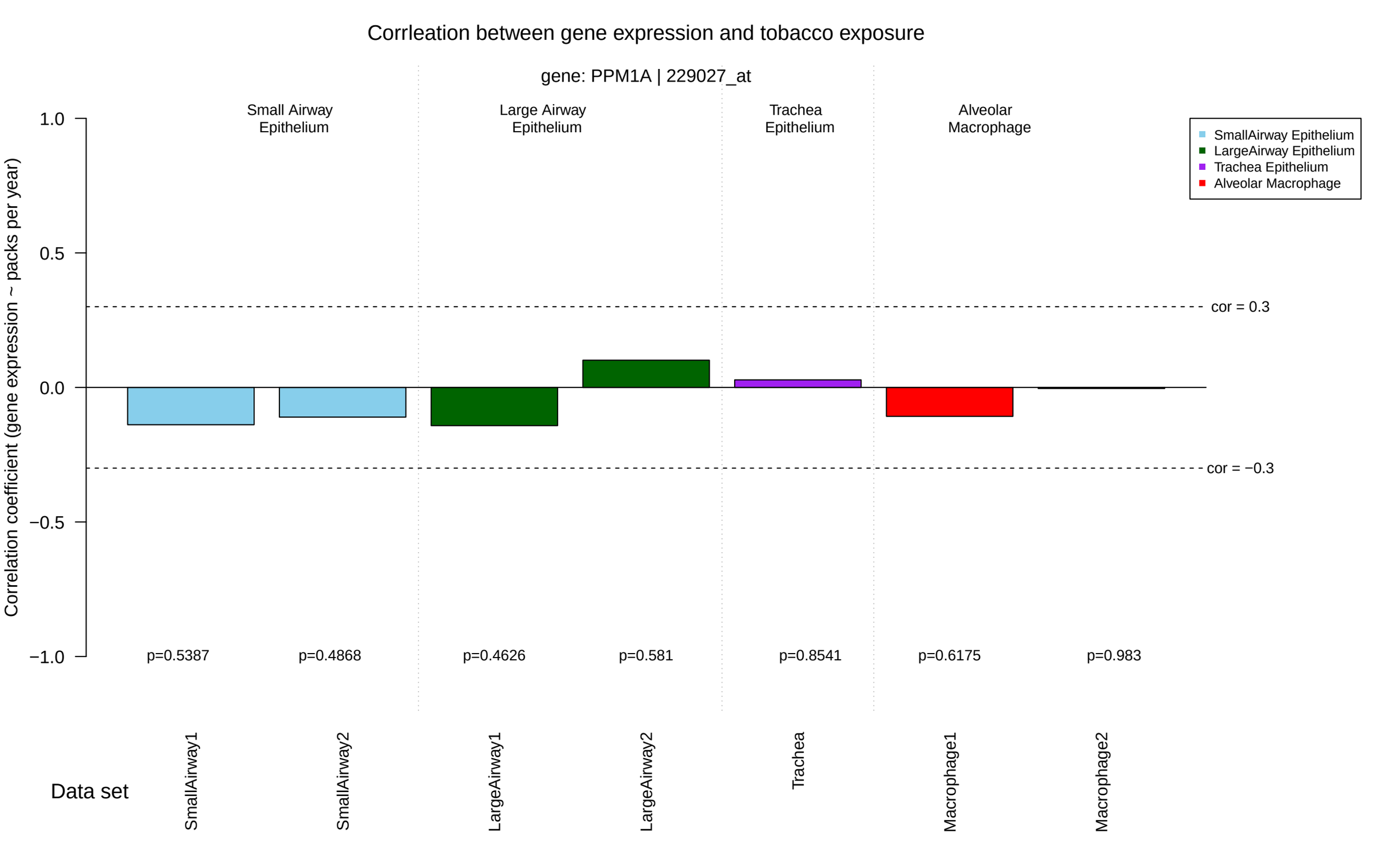

set: LargeAirway1 (p=0.0925)
PPM1A | 229027_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5336)
PPM1A | 229027_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9131)
PPM1A | 229027_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3071)
PPM1A | 229027_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4819)
PPM1A | 229027_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PPM1A | 229027_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5387
p=0.4868
p=0.4626
p=0.581
p=0.8541
p=0.6175
p=0.983
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2